

# Decoder-only conditional latent models with temporal mixture priors enable zero-shot cross-lab mouse sleep staging and substage discovery

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## Abstract

Manual scoring of mouse sleep from electroencephalography (EEG) and electromyography (EMG) lacks consensus across laboratories, and supervised deep learning models trained at one site often fail on data from others until retrained on multi-laboratory labels. We present an unsupervised computational framework for population-level sleep staging on the Mouse Sleep Staging Validation dataset (MSSV; OpenNeuro ds006366), comparing a controlled model ladder on identical features and evaluation protocols: feature-domain hidden Markov model (HMM), HMM–Gaussian mixture variational autoencoder (HMMGMVAE) without subject conditioning, decoder-only conditional GMVAE (cGMVAE), and decoder-only conditional HMM–GMVAE (cHMM–GMVAE) with a warm sticky HMM–Gaussian mixture model (GMM) latent prior. Models are trained on leave-mice-out cross-validation folds pooling laboratories 2, 3, and 5. Held-out mice are staged by zero-shot inference: spectral epochs are encoded without subject identifiers and states are assigned from the learned mixture prior only, without per-mouse manual calibration. On joint cross-lab holdout, cHMM–GMVAE achieves mean best-of-three prior normalized mutual information (NMI) 0.56 across four folds (range 0.38–0.65), compared with 0.33 for matched cGMVAE and 0.51 for a feature HMM baseline. A substage resolution sweep and physiology panels refine wake, NREM, and REM structure and reveal transition substates at macro boundaries. These results show that decoder-only conditioning with temporal mixture priors improves unsupervised generalization across mice and acquisition sites while supporting biologically interpretable substage discovery.

## Author summary

We study how *Mus musculus* sleep can be classified automatically from brain and muscle signals when recordings come from different laboratories with different equipment. Expert human scorers disagree across labs, and supervised computer models often fail on new sites. We built a family of unsupervised models that learn shared sleep patterns from many mice, then test them on held-out animals they have never seen—without asking a human to calibrate each new recording. Our best model combines a neural network that reads EEG and EMG with a temporal prior that enforces realistic sleep bout structure. It outperforms simpler models on cross-lab holdout and discovers finer sleep substates whose brain-wave profiles match known

physiology. We validate substages with power spectra, muscle activity, and bout lengths, with guidance from sleep neuroscientists. This work offers a label-free route to comparing sleep across studies and may help researchers find transition states between wake, NREM, and REM that are hard to score by eye alone.

# Introduction

Rodent sleep is scored from EEG and EMG into wake, non-rapid eye movement (NREM), and rapid eye movement (REM), yet montage and laboratory protocols differ across studies [1]. Rose et al. show that manual scoring lacks cross-lab consensus and that published deep models generalize poorly across MSSV sites until retrained on multi-laboratory data [2], motivating label-agnostic staging and rigorous holdout evaluation. Supervised models such as SPINDLE and REST achieve high cross-lab accuracy but require expert labels [3, 4]. Unsupervised tools include classical FASTER clustering [5], FASTER2 Gaussian HMMs on hand-crafted features [6], SegWay’s per-mouse k-means plus HMM smoothing [7], mcRBM substage discovery [8], recent WUCSS light/deep NREM states [9], and transition-focused supervised classifiers [10]; none combine deep latent learning, sticky HMM–GMM priors, and population zero-shot holdout on MSSV.

Our MSc thesis showed that subject-conditioned GMVAEs improve population staging ( $\text{NMI} \approx 0.70$ ) over feature HMMs ( $\approx 0.51$ ) [11], with interpretable substages but encoder conditioning that leaks subject identity at evaluation. We make three contributions: (i) decoder-only conditional variational autoencoder (cVAE) training with zero-shot holdout inference (no subject embeddings at staging); (ii) a fair model ladder ending in cHMM–GMVAE with temporal mixture priors [12–14]; and (iii) substage biology with expert validation. Synthetic validation, encoder+decoder ablations, and full  $K=13$  taxonomy are in S4 Fig and S1 Appendix.

## Results

### Cross-lab holdout ladder

We evaluated decoder-only conditional models under joint population training with leave-mice-out four-fold cross-validation across MSSV laboratories 2, 3, and 5 (Fig 1). Training uses subject embeddings in the decoder only; holdout staging encodes EEG/EMG without subject identifiers and assigns states from the learned HMM–GMM prior (Fig 1B), contrasting with per-mouse calibration in AccuSleep [15]. Fig 2 summarizes prior NMI on held-out mice; Table 1 lists per-fold values.

**Fig 1. Study overview and zero-shot holdout protocol.** **A**, MSSV multi-laboratory cohort with four-fold cross-validation; mice from three folds train a population model evaluated on the held-out fourth fold. **B**, Training uses decoder-only subject conditioning for reconstruction; holdout staging encodes EEG/EMG without subject identifiers and assigns states from the HMM–GMM prior only. **C**, Model ladder from feature HMM through HMMGMVAE, cGMVAE, and cHMM–GMVAE.

Across four completed joint folds, cHMM–GMVAE exceeded cGMVAE on every fold (mean prior NMI 0.56 vs. 0.33;  $\Delta = +0.22$ ). The gain was largest on folds 1 and 4 (0.25 and 0.35 NMI) and smallest on fold 2 (0.11), where both models fell below the thesis feature HMM reference (0.51). Within-lab holdout revealed laboratory heterogeneity: cHMM–GMVAE led on laboratory 3 (up to 0.76 NMI) while cGMVAE remained competitive on laboratory 2 (S2 Fig, S2 Table). HMMGMVAE locked holdout runs are in progress; the full four-rung ladder including per-fold breakdown is in S1 Fig.

**Fig 2. Joint cross-lab holdout prior NMI.** **A**, Mean best-of-three prior NMI across four folds for cGMVAE and cHMM–GMVAE; error bars span fold min–max; dashed line marks thesis feature HMM baseline (0.51). **B**, Per-fold grouped bars (labs 2, 3, and 5 pooled).

### Substages and transitions

We swept mixture component count  $K \in \{3, 5, 7, 9, 11, 13, 15\}$  on locked cHMM–GMVAE (joint fold 4) and selected  $K$  by validation score and cross-seed stability (S3 Fig). Fig 3 shows a physiology grid for the ten most occupied substages (EEG/EMG power spectral density,  $\theta/\delta$  ratio, EMG power, bout length, macro-label composition, subject mixing), following the mcRBM paradigm [8]. Fig 4 shows the substage transition matrix and a two-hour hypnogram excerpt aligned to expert macro labels; transition-rich states

**Table 1. Joint holdout prior NMI by fold (best-of-three seeds, locked recipes).**

| Fold        | cGMVAE | cHMM-GMVAE | HMM <sup>†</sup> |
|-------------|--------|------------|------------------|
| 1           | 0.361  | 0.612      | 0.51             |
| 2           | 0.275  | 0.381      | 0.51             |
| 3           | 0.407  | 0.594      | 0.51             |
| 4           | 0.296  | 0.646      | 0.51             |
| <b>Mean</b> | 0.335  | 0.558      | 0.51             |

<sup>†</sup>Thesis population leave-one-subject-out reference (feature HMM), not re-estimated per fold.

concentrate at wake-NREM and NREM-REM boundaries [10, 16]. Expert substage names and a curated summary table will be added after neuroscientist review (S1 Appendix).

**Fig 3. Substage physiology panels.** Rows: top-occupancy discovered substates; columns: EEG PSD, EMG PSD,  $\theta/\delta$ , EMG power, bout length, macro-label mix, subject mixing. Holdout cHMM-GMVAE, joint fold 4, seed 1.

**Fig 4. Substage dynamics.** **A**, Row-normalized transition matrix (probability of next state given a state change). **B**, Two-hour hypnogram excerpt: discovered substates (top) and expert macro labels (bottom).

## Discussion

Rose et al. establish that cross-lab supervised staging inherits rater disagreement [2]; our unsupervised ladder complements this by discovering structure without training on one laboratory’s label biases. Zero-shot holdout inference (encoder plus temporal prior, no subject identifier) contrasts with transductive calibration in AccuSleep and per-animal fitting in SegWay [7, 15]. Temporal mixture priors helped when bout structure and laboratory-specific spectra aligned with sequence modelling (laboratories 3 and 5); static GMM decoders sufficed on some montages (laboratory 2). Decoder-only conditioning is the main methodological advance over the thesis encoder+decoder setup for population holdout [11]. Absolute holdout NMI remains below curated thesis leave-one-subject-out performance; relative gains over the ladder and substage biology are the primary claims. Limitations include seed fragility on difficult folds, per-laboratory spectral front-end locks (not per-mouse manual scoring), and incomplete HMMGMVAE locked holdout runs.

## Materials and methods

### Animals and dataset

We used the MSSV dataset (OpenNeuro ds006366) [1]: 92 mice, laboratories 2, 3, and 5, 4 s epochs, expert wake/NREM/REM labels for evaluation only. All experiments used previously published data; no new animal procedures were performed in this study.

### Preprocessing and features

Per-laboratory spectral front-end locks were fixed from incohort tuning [11]. Wide multilayer perceptron encoder-decoder with latent dimension 6; convolutional encoder on spectral inputs.

## Model ladder

Locked fair-comparison recipes: feature HMM baseline (thesis); HMMGMVAE ( $emb=0$ , sequence length  $T=64$ , HMM-GMM prior); cGMVAE (decoder-only,  $emb=8$ , GMM prior); cHMM-GMVAE (+ warm sticky HMM-GMM,  $\kappa=0.92$ ) [12, 13, 17].

## Training

Population training on cross-validation training folds (joint: all laboratories; within-lab: single laboratory). Subject embeddings ( $emb=8$ ) were concatenated to the decoder only, not the encoder, during reconstruction and prior learning.

## Inference (holdout)

For held-out validation mice, sleep-state labels were obtained by encoding spectral epochs without subject identifiers, then assigning states via the learned GMM or sticky HMM-GMM prior on latent means (prior prediction). Subject embeddings were not used at holdout evaluation; the decoder was not invoked for staging.

## Evaluation

Leave-mice-out four-fold holdout; prior-prediction NMI versus expert macro labels; best of three random seeds per configuration.

## Substage sweep and physiology

Mixture component sweep  $K \in \{3, 5, 7, 9, 11, 13, 15\}$ ; physiology panels and transition matrices from holdout validation predictions.

## Data availability

MSSV is available on OpenNeuro (ds006366). Processed features and fold manifests are described in the SPA repository.

## Code availability

Training, evaluation, and figure scripts will be archived on Zenodo with a tagged release before publication (DOI pending).

## Acknowledgments

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## Supporting information

**S1 Fig. Per-fold holdout ladder.** Grouped bar chart of prior NMI for all four folds and four model rungs.

**S2 Fig. Within-laboratory holdout heatmap.** Laboratory  $\times$  model prior NMI for within-lab scope (do not compare absolute values to joint scope).

**S3 Fig. Substage resolution sweep.** Validation score versus mixture component count  $K$  on locked cHMM-GMVAE (joint fold 4).

|                                                                                                                                                                                |           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>S4 Fig. Thesis cGMVAE <math>K=13</math> taxonomy.</b> Full PCA, transition graph, and Fig-27-style panel from the MSc thesis.                                               | 91<br>92  |
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